

Harshavardan Chappidi

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with **6+ years** of experience designing large-scale data platforms across **AWS, Azure, and hybrid cloud environments**. Skilled in building reliable **batch** and **streaming pipelines** using **Spark, Databricks, Kafka, Delta Lake**, and modern data stack tools including **dbt** and **Airflow**. Strong track record in optimizing data systems for **analytics, ML feature engineering, and real-time decision workflows**. Experienced in **CI/CD automation, SQL performance tuning, and GenAI-driven solutions** that improve developer efficiency. Known for partnering closely with **engineering, data science, and product teams** to deliver high-impact, scalable data solutions that support business growth.

TECHNICAL SKILLS

Programming Languages: Python, Java, Scala, R, SQL, Shell Scripting

Big Data & Distributed Computing: Apache Spark, PySpark, Hadoop (HDFS, MapReduce, Hive), EMR, Delta Lake, Apache Iceberg

Data Engineering & ETL Tools: Apache Airflow, AWS Glue, Azure Data Factory (ADF), Databricks, dbt Core, SSIS, Prefect

Data Streaming & Messaging: Apache Kafka, Apache Flink, Spark Structured Streaming, Azure Stream Analytics, gRPC

Machine Learning & AI: LLM Fine-tuning, Prompt Engineering, OpenAI APIs, GenAI Workflow Automation, Feature Store Development, Model Serving Pipelines, ML Monitoring, Feature Engineering, MLflow, SageMaker, Scikit-learn, Hyperparameter Tuning, Python (Pandas, NumPy), SQL Generation with LLMs

Cloud Platforms: AWS (S3, EC2, Lambda, Redshift), Azure (Databricks, Synapse, Event Hubs)

Databases & Data Warehousing: Snowflake, Redshift, Oracle, PostgreSQL, MySQL, SQL Server, Teradata, MongoDB

DevOps & MLOps: Docker, Kubernetes, Jenkins, GitLab CI, GitHub Actions, Terraform, CI/CD, Model Deployment Workflows

Data Visualization & Reporting: Tableau, Power BI, Amazon QuickSight, SSRS, Excel, Matplotlib

Methodologies: Agile, Scrum, Waterfall, DevOps

PROFESSIONAL EXPERIENCE

Amazon | WA | *Senior Data Engineer*

May 2025 – Present

- Designed and implemented a **petabyte-scale data lakehouse** on **AWS**, leveraging **Lake Formation, Iceberg, and Amazon Redshift** to integrate diverse data sources and enable unified analytics and machine learning across business units.
- Developed and optimized **distributed data processing pipelines** using **Apache Spark (AWS EMR)** and **AWS Step Functions**, improving data freshness for critical datasets by **20%** and reducing operational costs.
- Collaborated with **data science teams** to productionize **ML feature stores** and **model serving pipelines**, directly supporting high-impact **recommendation** and **personalization models** that improved customer engagement metrics.
- Built and maintained a **CI/CD framework for data pipelines** using **GitHub Actions** and **Apache Airflow**, automating DAG deployment, validation, and data quality checks, reducing manual work by **35%** and increasing pipeline reliability.
- Applied **Generative AI** techniques by fine-tuning a **large language model (LLM)** to generate and optimize **SQL queries** from natural language inputs, reducing analyst query development time by **20%**.

Microsoft | WA | *Data Engineer*

September 2023 – April 2025

- Designed and developed **scalable data pipelines** on **Azure Data Factory** and **Databricks**, automating ingestion and transformation of **10+ TB** of daily telemetry data, increasing data availability for analytics by **20%**.
- Built a **real-time data processing platform** using **Apache Kafka** and **Apache Spark Structured Streaming** on Databricks, reducing data latency for key business metrics by **30%** and enabling near real-time insights.
- Implemented **dbt Core** to modularize and automate data transformations, converting raw telemetry into **ML-ready feature tables**, improving data quality and cutting feature engineering time by **15%**.
- Architected and deployed **Power BI** dashboards integrated with **Azure Synapse Analytics**, enabling real-time monitoring of **ML model performance** and **feature drift**, helping data scientists iterate **10% faster**.
- Accelerated development of **Python data transformation scripts** and **PySpark jobs** by adopting **GitHub Copilot**, improving developer productivity by **15%** while maintaining quality through structured code reviews.
- Mentored and trained **3 junior data engineers** in modern data stack technologies, including **dbt, feature store design, and GenAI API integration**, accelerating their onboarding and enabling independent project contributions within **6 weeks**.

- Partnered with **risk analytics** and **trading teams** to design real-time data platforms that strengthened regulatory reporting and supported customer analytics in high-stakes **financial environments**.
- Developed **PySpark** and **SQL processing** frameworks on **Databricks** and **AWS EMR**, integrating **Delta Lake** for **ACID compliance**, schema validation, and **Kafka** streaming to reduce processing errors and improve consistency.
- Automated complex **ETL pipelines** using **Apache NiFi** and **Informatica**, integrating data from **DB2**, **SQL Server**, **REST APIs**, and **market feeds** while embedding quality checks and **audit logging**.
- Built **Python-based** data quality frameworks using **Pandas** and **PyTest**, adding custom validation rules that increased data reliability by **25%** and reduced manual checks by **10 hours per week**.
- Created **CI/CD pipelines** with **Jenkins**, **GitLab CI**, and **Terraform**, adding **Python automation** for infra provisioning and deployments, achieving a **15% reduction** in deployment time.
- Optimized **Teradata** and **Oracle** data warehouses using advanced **SQL techniques** (partitioning, indexing, query tuning), lowering report generation time by **20%** for business users.

Abbott | India | Junior Data Engineer**January 2018 – March 2020**

- Engineered **MongoDB data models** for unstructured clinical datasets, using schema optimization and sharding to boost real-time query performance by **25%**.
- Designed and deployed **Azure Data Factory** pipelines to orchestrate ETL workflows across **Azure Blob Storage** and **SQL Database**, improving data freshness for analytics by **40%**.
- Built and automated scalable data pipelines using **Azure Functions** and **Logic Apps** to process clinical trial datasets, reducing manual workload by **35%** and improving operational efficiency.
- Improved large-scale processing by developing and maintaining **MapReduce** jobs on **Hadoop** clusters, reducing batch execution times by **20%** for regulatory workloads.
- Translated business requirements into interactive **Power BI dashboards** using **advanced SQL** (window functions, CTEs), enabling **self-service analytics** and reducing reporting delays by **30%**.

EDUCATION

Master's in Business Administration (Data Analytics) | University of Wisconsin, Milwaukee, WI, USA.**Bachelor's in Computer Science and Engineering |** Vellore Institute of Technology, Tamil Nadu, India.